

Q1. Explain the difference between Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-batch Gradient Descent.

Also compare them in terms of speed, stability, and practical usage.

Q2. Why does Momentum help reduce oscillation and speed up learning?

Explain using the idea of zig-zag movement and velocity.

Q3. What is the main problem of AdaGrad?

Explain how it affects the learning rate during training.

Q4. How does RMSProp solve the problem of AdaGrad?

Explain the difference between full accumulation and moving average.

Q5. Why is Adam considered a combination of Momentum and RMSProp?

Explain in simple terms.

Q6. Given:

$w = 5$, gradient (dw) = 2, learning rate = 0.1

Calculate the updated weight using SGD.

Q7. Which optimizer would you choose for:

a) sparse data (like NLP tasks)

b) deep neural networks

Explain your choices.

Q8. If your model is learning very slowly, which method would you try:

Momentum, RMSProp, or Adam?

Explain your answer.